



BACHELOR OF INFORMATION TECHNOLOGY (HONS)

FINAL EXAMINATION
JUNE 2023

Course : EC3123 & EC3239 (Operating Systems Principles) Time: 8.00am – 11.00am
(3 hours)

Lecturer: Wan Nor Al-Ashekin Bt. Wan Husin

Date: 26 September 2023

Instructions:

Answer ALL questions in the Answer Booklet provided.

The maximum number of marks is 100.

This exam paper consists of 4 printed pages.
(excluding the front cover)

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO

1. a. Define how the operating system works. (6 MARKS)
- b. List any FIVE (5) resources of networked operating system. (5 MARKS)
- c. Explain THREE (3) tasks that performed by the operating system to support the system components. (6 MARKS)
- d. Given the model of non-networks operating system in Figure 1: (8 MARKS)

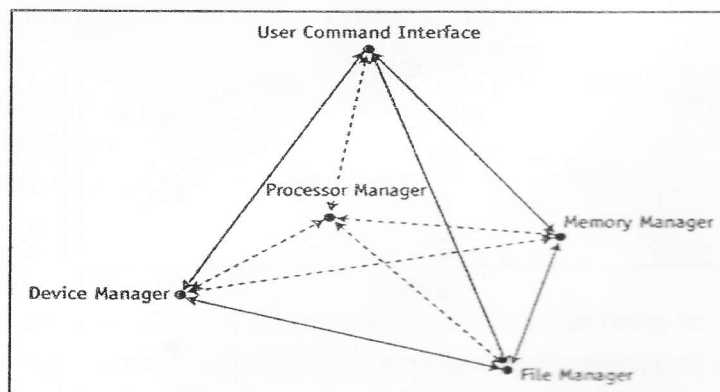


Figure 1

Describe FOUR (4) tasks that each of the sub-system has to perform.

2. a. Explain THREE (3) types of scheduler in operating system. (9 MARKS)
- b. Given the following processes and burst time in Table 1.

Table 1

Process	Burst Time	Priority
P1	10	2
P2	15	1
P3	20	3
P4	25	4

- i. Draw a Gantt Chart for Priority scheduling algorithm. (2 MARKS)
- ii. Draw a Gantt Chart for Round Robin scheduling algorithm with the Time Quantum = 8. (2 MARKS)

(4 MARKS)

- c. Write the Turnaround Time of the scheduling algorithms in Question 2(b)(i) and Question 2(b)(ii).
- d. Discuss any TWO (2) benefits of multithreading in operating system. (4 MARKS)
- e. (4 MARKS)

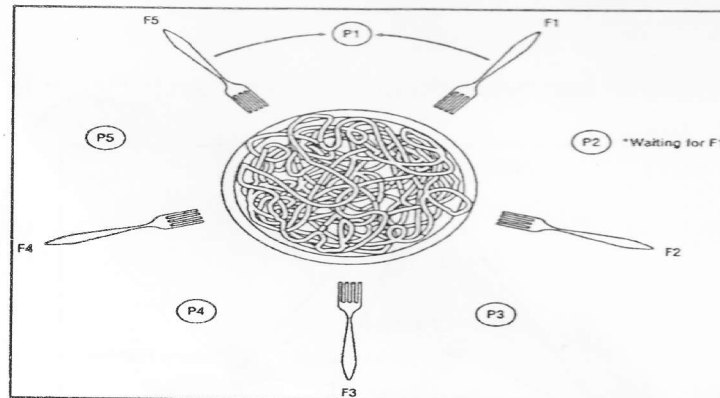


Figure 2

Discuss the Starvation Philosophy shown in Figure 2.

3. a. Write about the concept of paging and it's implementation in the memory. (6 MARKS)
- b. Illustrate a figure that show the shared pages for three processes; P1, P2 and P3 in memory management. (7 MARKS)
- c. Given the following information on users, files and permission modes in Table 1:

Table 1

User	Directory Name	File Name				
		File_1	File_2	File_3	File_4	File_5
Sonia	EC3122	Read	Read Write	Read Write	Owner Read Write Execute	Read Write
Jaya	EC3127	Read Write	Owner Read Write Execute	Read Write	Read	Read Write
Nisa	EC3120	Owner Read Write Execute	Read	Owner Read Write Execute	Read	Read Write

The user's home directory as follows:

- /home/Sonia
- /home/Jaya
- /home/Nisa

Based on the information given Table 1, draw the hierarchical/ tree structure including the user's home directories, directories and files.

(12 MARKS)

4. a. Define the function of following commands of Ubuntu Linux operating system:

(5 MARKS)

- pwd
- whoami
- man login
- sudo su -
- rm -r

- b. Write a single Ubuntu Linux command in octal form to set the following permissions for the file name *assignment*:

(8 MARKS)

- rwx rwx rwx
- rwx r—r—
- rw- rw- rw-
- rw- --- ---
- rwx r-w ---
- rwx --- ---
- rwx rw—r—
- rw- rw- r--

- c. Assume that the current working directory is /home/wanashekin. Write a single Linux commands for the following statements:

- User Wanashekin wants to remove two empty directories called *EC2339_Final Examination* and *EC3123_Lab*. (1 MARK)
- User Wanashekin wants to create a directory called *MyDiary*. (1 MARK)
- User Wanashekin wants to create a file called *Monthly_Activity* in directory *MyDiary*. (1 MARK)
- User Wanashekin wants to create a file called *My_Assignment* with some data in directory */home/wanashekin*. (1 MARK)
- User Wanashekin wants to view the list of files and file permissions in directory */home/wanashekin*. (1 MARK)
- User Wanashekin wants to add a new user with user's home directory called *Maria* in the system. (1 MARK)
- User Wanashekin wants to remove user *Maria* from the system. (1 MARK)

- viii. User Wanashekin wants to add a new group called *Junior* with primary group ID 2100. (1 MARK)
- ix. User Wanashekin wants to create a text file called *Event.txt*. (1 MARK)
- x. User Wanashekin wants to view the list of files in directory /tmp from home directory. (1 MARK)
- xi. User Wanashekin wants to move house3 and house4 to the \$HOME/projects/houses/doors directory. (1 MARK)
- xii. User Wanashekin want to create nine empty files (house1 to house9) in directory projects (1 MARK)

-END OF QUESTION PAPER-